

WELLSVILLE, NY, USA

WMO: 725157

Lat: 42.109N

Lon: 77.992W

Elev: 647

StdP: 93.79

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54757

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -18.2	(c) -16.0	(d) -23.4	(e) 0.5	(f) -17.0	(g) -20.9	(h) 0.6	(i) -14.8	(j) 12.0	(k) -5.4	(l) 11.2	(m) -5.7	(n) 4.6	(o) 300	(p) 0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.0	(c) 28.9	(d) 20.7	(e) 27.6	(f) 20.1	(g) 26.3	(h) 19.4	(i) 22.3	(j) 26.6	(k) 21.5	(l) 25.6	(m) 20.7	(n) 24.7	(o) 4.0	(p) 260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.0	17.0	24.6	20.1	16.0	23.7	19.2	15.1	22.9	68.6	26.5	65.3	25.4	62.5	24.8	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.7	(b) 9.1	(c) 8.2	(e) -21.4	(f) 31.1	(g) 2.8	(h) 1.8	(i) -23.4	(j) 32.4	(k) -25.1	(l) 33.5	(m) -26.6	(n) 34.5	(o) -28.6	(p) 35.8		
(a) 10.7	(b) 9.1	(c) 8.2	(e) -21.8	(f) 23.7	(g) 2.7	(h) 0.9	(i) -23.8	(j) 24.3	(k) -25.3	(l) 24.8	(m) -26.9	(n) 25.3	(o) -28.8	(p) 25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-5.0	-4.1	0.4	7.4	13.6	17.6	20.1	19.3	15.8	9.7	3.7	-1.7
	DBStd	10.27	6.85	6.17	6.55	5.78	5.06	3.61	3.04	3.01	3.98	5.23	5.41	5.46
	HDD10.0	1946	467	397	304	118	22	1	0	0	2	72	199	363
	HDD18.3	3929	725	629	554	332	162	56	17	26	98	272	439	620
	CDD10.0	1262	1	1	8	38	134	228	314	288	176	62	9	2
	CDD18.3	203	0	0	0	2	15	34	73	55	21	3	0	0
	CDH23.3	1267	0	0	0	23	117	215	460	306	133	12	0	0
	CDH26.7	217	0	0	0	2	16	33	94	49	23	1	0	0
Wind	WSAvg	3.8	4.6	4.6	4.5	4.3	3.5	3.1	2.8	2.6	2.8	3.7	4.1	4.5
Precipitation	PrecAvg	990	66	49	73	80	82	102	105	102	99	88	80	69
	PrecMax	1237	107	107	125	130	168	170	212	196	173	177	135	121
	PrecMin	747	19	20	37	43	28	10	45	49	52	29	27	40
	PrecStd	131	24	23	22	24	37	43	44	38	36	41	29	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	16.2	20.8	26.4	28.4	29.4	31.1	29.9	29.4	24.9	19.5	15.4
		MCWB	11.4	11.8	13.0	14.5	18.9	21.0	21.2	21.8	21.1	18.2	12.9	12.5
	2%	DB	11.0	11.1	16.4	22.9	26.6	27.6	29.0	28.0	26.7	22.4	16.2	11.5
		MCWB	8.7	7.7	10.2	13.2	17.7	20.4	21.1	20.4	19.7	16.3	11.3	9.2
	5%	DB	7.6	7.6	13.2	19.9	24.3	26.0	27.5	26.6	24.8	20.2	14.0	8.6
		MCWB	5.2	4.1	9.1	11.7	16.8	19.4	20.7	19.8	18.7	15.8	10.0	6.4
	10%	DB	4.7	4.6	10.5	17.2	22.1	24.1	26.0	25.0	22.6	17.7	12.3	6.0
		MCWB	2.7	2.0	6.4	10.4	15.4	18.2	19.6	19.1	17.7	13.9	8.9	3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	12.3	14.0	16.7	20.8	22.5	23.5	23.1	22.8	20.2	14.9	12.7
		MCDB	13.6	15.4	19.6	23.3	26.6	27.1	28.3	27.8	26.9	23.0	17.4	15.0
	2%	WB	9.0	8.0	11.8	14.7	19.3	21.2	22.4	21.9	20.9	18.2	12.5	9.4
		MCDB	10.5	10.1	14.9	20.3	24.3	26.0	27.0	25.6	25.4	20.9	14.6	11.1
	5%	WB	5.6	5.0	9.7	13.0	18.0	20.2	21.6	21.1	19.8	16.4	10.7	6.6
		MCDB	7.3	7.3	12.9	18.1	22.2	24.5	25.7	24.9	23.4	19.2	13.5	8.3
	10%	WB	2.6	2.0	6.7	11.4	16.8	19.4	20.7	20.1	18.6	14.2	9.0	4.0
		MCDB	4.6	4.5	10.2	16.2	20.7	23.0	24.4	23.7	21.7	17.3	11.8	5.7
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.9	9.0	10.7	10.7	9.9	10.0	9.8	9.9	8.5	7.5	6.1
		MCDBR	9.5	10.8	12.6	14.5	12.4	11.3	11.0	11.1	11.5	11.0	10.4	9.3
		MCWBR	8.4	8.6	7.9	8.0	6.1	6.0	5.3	5.3	5.7	6.2	7.2	7.9
	5% WB	MCDBR	9.6	10.8	11.2	12.5	10.7	9.6	9.3	9.1	9.6	9.6	9.2	9.1
		MCWBR	9.2	9.4	8.2	8.1	5.8	5.7	5.1	4.9	5.3	6.4	7.5	8.5
Clear-Sky Solar Irradiance	taub	0.289	0.306	0.326	0.354	0.379	0.421	0.421	0.413	0.353	0.339	0.300	0.282	
	taud	2.381	2.332	2.382	2.367	2.325	2.225	2.229	2.242	2.403	2.450	2.525	2.473	
	Ebn at Noon	858	897	921	914	895	855	850	845	880	850	847	841	
	Edh at Noon	80	99	107	117	125	139	137	131	104	88	70	68	
All-Sky Solar Radiation	RadAvg	1.41	2.24	3.32	4.40	5.19	5.58	5.67	5.08	4.09	2.58	1.66	1.14	
	RadStd	0.12	0.26	0.34	0.48	0.53	0.47	0.44	0.31	0.44	0.27	0.23	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+85	N/A

Nomenclature: See separate page